

Revised Load Interconnection System Impact Study Report

For Queue Project Q17-0756

Gordonsville 138 kV
June 17, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0756. The Interconnection Customer proposes a 193.3 MW large load (data center) facility interconnecting at the Gordonsville 138 kV substation in the Virginia area. The request was received on March 10, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 316 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,265,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gordonsville to Gordonsville Energy Lp 138.0 kV	Louisa Generation Facility Substation 345/138kV XFMR	137.1	240	48.0
P.T.S to Medora 345.0 kV	Allegheny Energy Units 3 4 & 5 to Sugar Hill 345.0 kV	104.6	1,894	5.4
Handsome Lake to Ganso 500.0 kV L	Lions Rd to Piney Creek Project 500.0 kV line	103.5	2,387	6.2
Omni-Hackberry to Norwich 500.0 kV	Beaver to Trego 500.0 kV	102.6	2,332	7.1
Trego to Beaver 500.0 kV	Adams to Onancock Diesel Generation Facility 230.0 kV	102.0	3,123	8.5
Sugar Hill to Allegheny Energy Units 3 4 & 5 345.0 kV Line	Hilltop to Piney Crk Project 500.0 kV	100.3	1,785	6.5
Allegheny Energy Units 3 4 & 5 Substation 345/1kV Transformer Bank	Hilltop to Piney Creek Project 500.0 kV L	100.2	1,626	6.4
Allegheny Energy Units 3 4 & 5 500 kV/1 kV Transformer	Hilltop Substation to Piney Crk Project 500.0 kV	100.2	1,626	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gordonsville Energy Lp to Gordonsville 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,249,000	30% (\$1,848,000)
P.T.S to Medora 345.0 kV	Reconductor line	\$14,470,000	2% (\$268,000)
Ganso to Handsome Lake 500.0 kV Line	Reconductor line	\$39,089,000	1% (\$512,000)
Omni-Hackberry to Norwich 500.0 kV	Reconductor line	\$27,279,000	1% (\$306,000)
Trego to Beaver 500.0 kV	Reconductor line	\$144,385,000	1% (\$1,637,000)
Allegheny Energy Units 3 4 & 5 to Sugar Hill 345.0 kV	Reconductor line	\$6,475,000	1% (\$75,000)
Allegheny Energy Units 3 4 & 5 Substation 345/1kV Autotransformer	Replace transformer with higher-rated unit	\$18,051,000	1% (\$208,000)
Allegheny Energy Units 3 4 & 5 500/1kV Transformer	Replace transformer with higher-rated unit	\$35,703,000	1% (\$411,000)
	Total allocated to project		\$5,265,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.